Multiview Feature Learning and Multilevel Information Fusion for Joint Classification of Hyperspectral and LiDAR Data

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***Abstract*— With the rapid development of earth observation technology, the fusion of multisource remote sensing (RS) data has become an important research field in land cover classification. Particularly, the fusion of hyperspectral image (HSI) and light detection and ranging (LiDAR) data, which can provide complementary information to each other, has attracted an increasing attention for land cover classification tasks. However, most of the existing methods focus on multisource information fusion on feature level and ignore the discrepancy of the land covers with the same label among different kinds of viewpoints, which limit the further improvement of classification performance. So, this article proposes a novel HSI-LiDAR classification method based on multiview feature learning (MVFL) and multilevel information fusion (MLIF), which consists of a multiview data representation (MVDR) strategy, a multibranch dual-channel graph convolutional networks (MB-DCGCNs) model, and a progressively high-confidence label assignment (PHCLA) scheme. MVDR is designed by jointly utilizing multiple attributes of spatial information and multisource RS data to re-express multisource land covers with diversity and complementarity. The MB-DCGCNs model, which aims at integrating spectral–spatial–elevation information on feature level, is employed on the above multiview re-expressed HSI-LiDAR (MVR-HL) data to explore land cover feature representations from different views. Furthermore, a PHCLA scheme is proposed to classify the land covers with high-reliability by a combination of decision-level label prediction and pixel-level label assignment. Comparative experiments on two benchmark datasets with several state-of-the-art classification methods validate the effectiveness and superior performance of the proposed method.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image (HSI), light detection and ranging (LiDAR), multilevel information fusion (MLIF), multisource data classification.**

I. Introduction

LAND cover classification has always been a hot spot topic in the field of earth observation and monitoring, which provides constructive and valuable suggestions for various application aspects, such as crop production and management, urban planning, and mineral exploration. With the rapid development of remote sensor technology [1], remote sensing (RS) data can provide continuous observation of the earth surface and hence have become increasingly popular data resource for land cover classification. Because of the different imaging mechanisms, each kind of RS data has its own characteristics and advantages as well as limitations [2]. Specifically, hyperspectral images (HSIs) provide abundant and contiguous spectral information of ground objects, which facilitates the process of land cover classification. However, due to the lack of elevation information about different categories of land covers, the classification methods, which merely based on single HSI data, inevitably meet the performance bottleneck [3]. As shown in Fig. 1, different types of land covers, which contain similar material components, probably show similar spectral curves but possess clear difference in elevation (e.g., road and commercial). Meanwhile, the observed targets that belong to the same class (e.g., healthy grass) may present the same height but with totally different spectral reflectance curves in some complex imaging environment, such as cloud covering. Meanwhile, light detection and ranging (LiDAR) data characterize the 3-D structure of the surveyed target and contain detailed height information [4]. Owing to the diversities and complementarities of HSI and LiDAR data in information presentation, the collaborative classification of HSI-LiDAR data has been playing an increasingly active role in the research field of earth observation [5]. Meanwhile, the difference and heterogeneity that exist in the multisource data have also posed a great challenge for practical application.

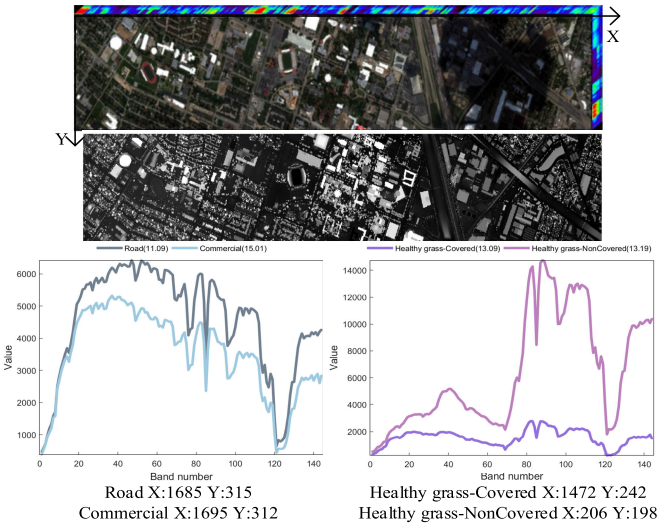


Fig. 1. Detailed information of land covers in the multisource data.

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In the past few decades, enormous efforts have been made to effectively integrate the respective advantages of HSI and LiDAR data and mine useful information for better land cover feature expression and classification performance. Considering that different types of features can describe the same object from different viewpoints, a variety of studies focus on how to properly realize multiple feature extraction and fusion for HSI-LiDAR collaborative classification. In the early studies, the researchers usually stacked various features, which extracted by different methods, including minimum noise fraction rotation (MNF), normalized difference vegetation index (NDVI), gray-level co-occurrence matrix (GLCM), and morphological attribute profiles (MAPs), and then utilized the typical classifiers, such as support vector machine (SVM) and random forest (RF) to complete the classification tasks [6], [7], [8]. Rasti et al. [9] used the orthogonal total variation component analysis (OTVCA) to estimate the fused information, which contain spectral information, spatial, and elevation information, which extracted from HSI-LiDAR data by extinction profiles (EPs) in a lower dimensional subspace for producing the final classification map. In [10], a feature fusion method named relative total variation structure analysis (RTVSA) is designed to remove useless spatial information derived from HSI and LiDAR data, which extracted by extended multiattribute profile (EMAP), Gabor filter, and local binary pattern (LBP). Jia et al. [11] proposed the multiple feature-based superpixel-level decision fusion (MFSuDF) method for the joint classification of HSI and LiDAR data, which contains entropy rate superpixel segmentation (ERS)-guided kernel principal component analysis (KPCA), multisource data feature extraction by 2-D and 3-D Gabor filters, and a decision-level fusion framework. In addition, graph-based fusion techniques and low-rank models were introduced to enhance multisource classification performance. Liao et al. [12] constructed a graph-based fusion scheme, in which the edges of the generated fusion graph were weighted by the distance between the stacked feature points. Gu and Wang [13] proposed a discriminative graph-based method, which couples dimension reduction and heterogeneous feature fusion from HSI and LiDAR data for urban area classification. Duan et al. [14] proposed a fusion approach based on multilevel structure extraction (MSE). First, the MSE is constructed and employed to extract the spatial and semantic information from original data. Then, a low-rank representation model is adopted to fuse the integrate the extracted features and the final labels are decided by maximum a posteriori estimation model. Tu et al. [15] defined homogeneous region fusion between the principal components (PCs) of the HSI and LiDAR data as a mathematical optimization problem and introduced the gradient transfer model to fuse spectral and digital surface model (DSM) information from various superpixel blocks for HSI and LiDAR data classification.

In most recent years, deep learning (DL) models that show powerful feature learning ability have attracted extensive attention from the research community [16]. Specifically, inspired by the encouraging classification performance of single-source RS data based on DL models [17], an increasing number of DL-based methods have emerged in the field of multisource data fusion and classification [18], [19]. As a representative DL structure, convolutional neural network (CNN) is an early and commonly used model to fuse HSI-LiDAR data for classification tasks. Ghamisi et al. [20] used traditional feature extraction method EP for spatial and elevation information extraction from both LiDAR and hyperspectral data. Then, CNN was explored for the first time to learn the higher level representations and ultimately produce the classification map. Hereafter, a series of two-tunnel CNN models have been designed to learn the characteristics of HSI and LiDAR data for more accurate identifications of ground objects, in which one CNN tunnel is used for hyperspectral feature extraction and another is introduced into LiDAR data [21], [22]. Afterward, some variants of CNN structure for multisource data fusion and classification have been gradually proposed to achieve more accurate classification results. For instance, Zhao et al. [23] proposed a hierarchical random walk network (HRWN), which can jointly optimize dual-tunnel CNN and pixelwise affinity via a novel random walk layer, which enforced spatial consistency in the deeper layers of the network to explore deep features for multisource data classification. Hong et al. [24] developed two novel multimodal DL (MDL) frameworks for multisource RS classification using an encoder–decoder fusion module, which contain pixel-level labeling guided by a fully connected networks (FC-Nets) and spatial–spectral joint classification with CNNs-dominated architecture. Lu et al. [25] proposed a coupled adversarial learning-based classification (CALC) method, which is based on a coupled adversarial feature learning (CAFL) subnetwork and a multilevel feature fusion classification (MFFC) subnetwork to perform multisource data fusion and classification. On the one hand, the CAFL subnetwork establishes an adversarial game between dual generators and discriminators for joint learning of the highly similar high-level semantic features and different low-level detail features through weight sharing and linearly weighting the encoded features of the input HSI and LiDAR data. On the other hand, the MFFC performs decision fusion and classification on the learned multilevel joint encoding features.

In order to implement deep characteristics exploration from multisource data collaboratively and automatically, some groundbreaking DL frameworks have also been developed for HSI-LiDAR classification tasks. Wu et al. [26] developed a CNN-dominated multisource RS data classification backbone framework with a plug-and-play cross-channel reconstruction (CCR) module, which aims at learning the multimodal features in a more compact way. Zhao et al. [27] built a fractional Gabor convolutional network (FGCN), which uses octave convolution layers to perform multisource information fusion and preserve discriminative information, and furthermore, used the FGC layers to extract multiscale, multidirectional, and semantic change features. Hong et al. [28] proposed a deep encoder–decoder network (EndNet) architecture for the classification of HSI and LiDAR data, which fuses the multimodal information by enforcing the fused features to reconstruct the multimodal input in turn for better activating the neurons across modalities. Ding et al. [29] designed a novel global–local transformer network (GLT-Net) to combine local

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spatial convolutional encoding features with global spectral sequence encoding for promoting the improvement of classification results. The proposed GLT-Net method mainly consists of multiscale local spatial feature learning, the multiscale aggregated CNN for local spatial feature extraction of HSI and LiDAR data, a multihead attention mechanism for global spectral feature learning, and adaptive weighted decision-making based on the class probability estimation of local spatial features and global spectral features.

In summary, multisource data provide complementary information for describing the same-class land covers, which is helpful for distinguishing easily confused ground objects, which is expressed by the single-source RS data form. However, current HSI-LiDAR classification methods still face the following limitations and weakness. First, most of the existing works that adopt simple feature stacking or cascading may easily lead to the curse of dimensionality and poor classification performance without considering the difference and complementarity among these multiple features. In addition, the DL-based multisource data fusion methods usually utilize the CNN as the base structure, which focus on local information exploration and could not effectively construct long-range topological relations among the same category of the land covers from a global perspective. Moreover, the abovementioned work that aims at mining multiple information with complementarity and otherness in the multimodal and heterogeneous RS data still lack of the redundant information removing in different views and information interaction across views. Also, these methods could not give full play to the consistency and relevance of the same-attribute information represented by the multisensor RS data. Last but not least, an overlooked problem is that multimodal information fusion at different levels could enhance the hidden relationship of the same ground object, which are vital for revealing the intrinsic characteristics of land covers from different classes, especially in some complex scenes.

II. Motivation and Contribution

Due to the diversity and difference of information representation, a ground object could be described from different views and the joint exploitation of different views would contribute to boosting the classification performance [30]. Hence, multiview learning-based classification methods have been developed and attracted much attention in recent years. Gao et al. [31] built a CNN-based multiple feature learning (CNN-MFL) architecture for the extraction of multiple HSI features by several CNN blocks which concurrently exploit the interactions of different spectral and spatial contextual information. Liu et al. [32] proposed a deep multiview learning (DMVL) method for HSI classification, which learns view-invariant features to greatly improve classification accuracy in the case of small samples. In [33], the two dual-channel CNN model is deigned to explore both uniform LBP and HSI data simultaneously and an ensemble of multiple CNN classifiers (EMCNN) with superpixel smoothing is also proposed for the task of HSI classification. Although some existing multiview learning strategies successfully extract multiview features for HSI classification, the joint representation and use of multiview feature is still a challenging problem in the field of multisource data classification due to the multimodal data heterogeneity and otherness. In addition, these methods that usually based on CNNs focus on local information by fixed-size square regions and lack of the correlation exploitation across different views. As the latest research production in the field of DL models, graph convolutional networks (GCNs) can effectively handle graph structure data by modeling long-range relations between samples [34]. Unlike CNNs, GCNs could exploit relationships between samples to explore inherent structural similarity and the HSI classification methods based on the combination of CNNs and GCNs have come to the fore. Hong et al. [35] proposed an end-to-end trainable network, which integrates the features extracted from CNNs and the minibatch GCNs (miniGCNs). In [36], a CNN-enhanced GCN network, which integrates the graph encoding process, was designed to perform feature learning on small-scale regular regions and large-scale irregular regions and generate complementary spectral–spatial features at pixel and superpixel levels, respectively. Liu et al. [37] generated multilevel graphs by successively merging adjacent regions in HSIs to model their spatial topologies in a multiscale hierarchical manner. Since these methods leverage the advantages of the CNNs and GCNs to model relations between the samples, which compensate for the deficiency of depending only on GCNs, however, they ignore to deeply learn implicit information from different views and comprehensively use multiview information, which may result in low-quality land cover feature representations. Hence, it would be valuable to develop a multisource RS land cover classification model based on the characteristics of GCNs and multiview learning to effectively reveal the intrinsic relationship of the land covers and learn the distinguishable and robust features for improving classification accuracy.

In this article, we proposed a multiview feature learning (MVFL) and multilevel information fusion (MLIF) strategy for the joint classification of HSI and LiDAR data. First, considering that different attributes of spatial information can describe multiple properties of the same ground object [38], the capacity in identifying different kinds of land covers could be correspondingly strengthened via organically utilizing the multiple spatial information, which inherent in the multisource RS data. Hence, various high-quality feature extractors, i.e., EMAP [39], Gabor [40], and GLCM [41], are used to discover multiattribute spatial characteristics from multisource data [42]. Then, each kind of these spatial information is explicitly concatenated with the original HSI-LiDAR data to construct the multiview data representation (MVDR) with complementarity and diversity. Subsequently, a dual-channel GCN (DCGCN) model is designed and applied on each view of the re-expressed multisource data for integrating spectral–elevation–spatial information in one organic whole and strengthening the semantic relatedness of the same-class land covers. Thus, a multibranch DCGCN (MB-DCGCN) model based on one-view-one-DCGCNs mechanism is designed to remove the redundant information in each specific view and learn land cover feature representations of different views. The feature-level information fusion facilitates the process of the shared information exploration

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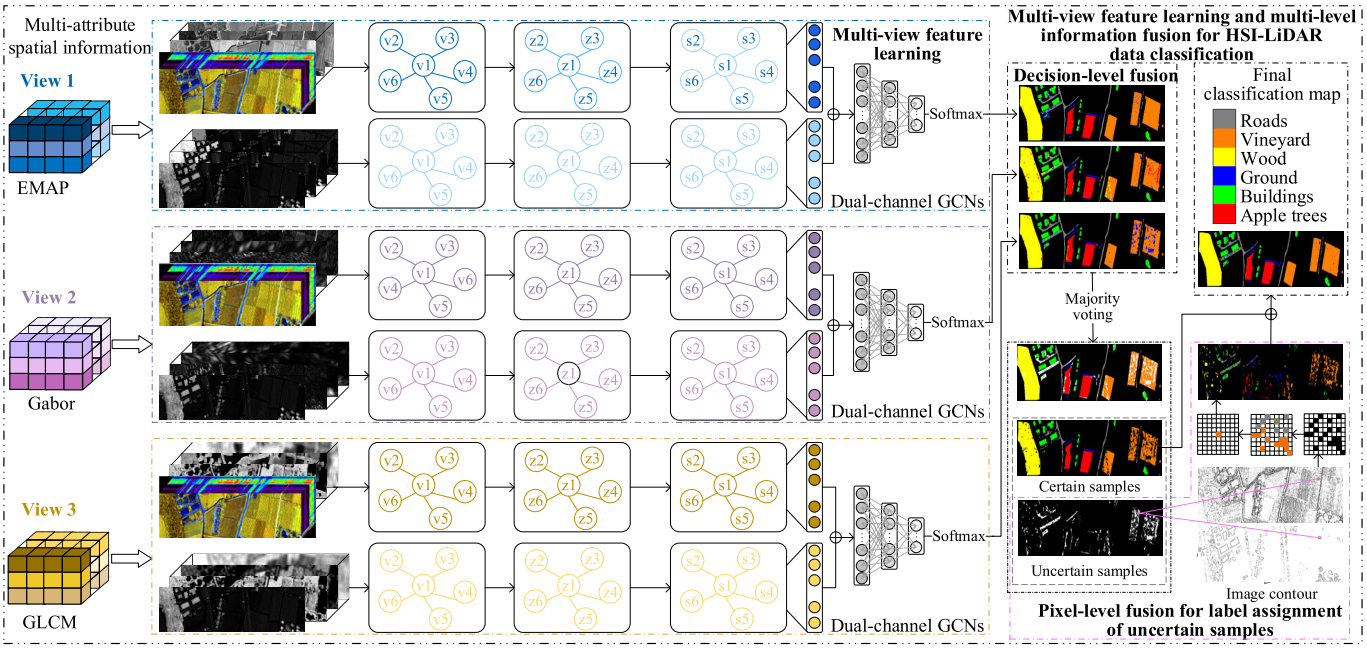


Fig. 2. Flowchart of the proposed method for the joint classification of HSI and LiDAR data. The variables of $\boldsymbol{V}$, $\boldsymbol{Z}$, and $\boldsymbol{S}$ denote vertexes, hidden representations via GCN layer, and hidden representations via ReLU layer, respectively.

among different views and the useless information elimination of each view simultaneously. Finally, a novel progressively high-confidence label assignment (PHCLA) scheme, which combined by decision-level label prediction and pixel-level label assignment, is proposed to yield the final classification result. The former introduces hard majority voting to fuse the primary output of individual branch in the MB-DCGCNs model and determines the class predictions of the samples, which share the same class label across views. The latter utilizes the semantic information alignment in the local region with homogeneity to realize label calibration for the residual samples [43]. The major contributions of this article can be summarized as follows.

1) First, multiattribute spatial features are utilized to reconstruct the multisource data with otherness and complementarity. After that, a MB-DCGCN model is developed to obtain the multiview feature representations, which contributes to discovering the underlying relationship in the same view and exploring the downstream complementarity and consistency among different views.
2) A hierarchical information fusion scheme is designed to effectively integrate the multimodal information from multisource and heterogeneous HSI-LiDAR data. In this way, the proposed method maximally preserves the hidden relevance among the same-class samples represented by multisensor RS data to overcome the within-class variations and between-class interference for guaranteeing the high reliability of the final classification result.

The rest of this article is organized as follows. A detailed description of the proposed method is presented in Section III. The experimental results and analysis are discussed in Section IV, followed by a conclusion in Section V.

III. Proposed Multisource Classification Framework

The framework of the proposed method for joint classification of HSI-LiDAR data is illustrated in Fig. 2. The proposed classification method is mainly composed of three parts, including the MVDR for multisource data, the MB-DCGCNs model, and the PHCLA scheme. Accordingly, we will elaborate each part in detail as follows.

A. MVDR for Multisource Data

It is well-known that there exits abundant and various information inherent in the multimodal and heterogeneous HSI-LiDAR data. Among a variety of information representations, different kinds of spatial information could reflect multiple spatial characteristics of land convers from different viewpoints, which motivates us to use different types of spatial feature extraction methods to fully exploit the spatial information from these two RS data. Considering that EMAP, Gabor, and GLCM could present complementary descriptions of the same object in the spatial domain (including geometry, directionality, size and randomness, and shape, respectively), we use these methods to extract diverse spatial properties of land covers from multisource data [38]. Then, inspired by multiview learning [30], [42], [43], [44], we integrate multiattribute spatial information with the HSI-LiDAR data to reconstruct multisource data representation with diversity and complementarity.

For clarity, given the original HSI dataset $\boldsymbol{H} \in \Re^{d_{\text{Spe}} \times N}$ of N samples with d_{Spe}-dimension spectral bands, which contains the training sample set $\boldsymbol{H}^{\text{tr}} = \{\boldsymbol{h}_i^{\text{tr}}\}_{i=1}^{N_{\text{tr}}} \in \Re^{d_{\text{Spe}} \times N_{\text{tr}}}$ and the testing sample set $\boldsymbol{H}^{\text{te}} \in \Re^{d_{\text{Spe}} \times N_{\text{te}}}$. Correspondingly, the LiDAR dataset can be denoted as $\boldsymbol{L} \in \Re^{d_{\text{Ele}} \times N}$, where d_{Ele} is the dimension of the elevation information. $\boldsymbol{Y}^{\text{tr}} = \{y_i^{\text{tr}}\}_{i=1}^{N_{\text{tr}}} \in \Re^{C \times N_{\text{tr}}}$ is a

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label vector corresponding to the training sample set and the samples are labeled from one to C. For the HSI data, the PC analysis (PCA) is first adopted to remove the redundant spectral information, and then, the first two PCs of the original HSI are used to produce multiattribute spatial information of the land covers [42].

1) EMAP Feature Extraction: Attribute profiles (APs) that originate from morphological profile can describe structural information by operating a series of thinning and thickening transformation on a grayscale image f, which can be shown as follows:

$$\begin{aligned}&\mathrm{AP}(f)\\&=\{\phi_n(f),\phi_{n-1}(f),\ldots\phi_1(f),f,\gamma_1(f),\gamma_2(f),\ldots\gamma_n(f)\}\end{aligned}\tag{1}$$

where ϕ and γ are the thickening and thinning operators with n morphological attributes, respectively. Then, for the multiple-value HSI data, the extended attribute profiles (EAPs) can be acquired by computing APs on the first two PCs extracted from the HSI using one attribute filter. Finally, by the use of multiple attribute filters, including a, area of the regions; d, length of the diagonal of the box bounding the region; s, standard deviation and i, moment of inertia, different attributes of EAPs are integrated to obtain the EMAP features

$$\mathrm{EMAP}=\{\mathrm{EAP}_a,\mathrm{EAP}_d,\mathrm{EAP}_s,\mathrm{EAP}_i\}\tag{2}$$

where, for instance, EAP_a means calculating the EAP on each PC of the HSI with the attribute a. For the 2-D LiDAR data, the EMP would be calculated for spatial feature extraction of this specific view.

2) Gabor Feature Extraction: A Gabor filter bank, which contains a set of Gabor filters defined by a Gaussian function with different scales (spatial frequencies) and orientations, is employed on the first two PCs of the HSI and the original LiDAR data to generate multisource Gabor features. The Gabor filter that could be regarded as a product of a Gaussian function and a sinusoidal plane wave is given as follows:

$$\begin{aligned}g_{f,\theta}(x,y)&=\frac{1}{2\pi}\exp\left(-\left(\alpha^2x'^2+\beta^2y'^2\right)\right)\exp\left(j2\pi\omega x'\right)\\x'&=\left(x-\frac{m+1}{2}\right)\cos\theta+\left(y-\frac{n+1}{2}\right)\sin\theta\\y'&=\left(x-\frac{m+1}{2}\right)\sin\theta+\left(y-\frac{n+1}{2}\right)\cos\theta\end{aligned}\tag{3}$$

where (x,y) are the spatial variables, m and n represent the size of Gabor filters, ω denotes the wavelength of the sinusoid, θ is the rotation of Gaussian function, and α and β mean the sharpness of Gaussian function. The Gabor filter bank that constituted by variations in the parameters of (3) is used to detect borders or textures by utilizing different orientations and scales.

3) GLCM Feature Extraction: The GLCM provides the statistical information about the relative position of a pixel and its neighboring pixels in gray-scale images. For example, in a given $H\times W$ image f, the value of the pixel pairs (i,j) in the GLCM, and it specifies the number of times the pixel with a value i occurred horizontally adjacent to a pixel with value j can be determined as

$$P_{ij}=\sum_{x=1}^{H}\sum_{y=1}^{W}\begin{cases}1, & \text{if } f(x,y)=i\vee f(x+\Delta x,y+\Delta y)=j\\0, & \text{else}\end{cases}\tag{4}$$

where offset $(\Delta x,\Delta y)$ is the distance between pixel (x,y) and its neighboring pixels. Thus, the GLCM method could distinguish the difference between smooth pixels with close pixel values and coarse pixels with divergent pixel values by calculating the conditional probability density in a square window. Aimed at well adapting to multiple scales of various land covers in the same scene, the GLCM textures, including contrast, correlation, dissimilarity, angular second moment, entropy, uniformity, and inverse difference moment, are computed using different sizes of square windows to capture multiscale spatial information.

Different types of spatial information can reflect the spatial characteristics of the land covers from different points of view. Hence, multiple spatial feature extraction methods that mentioned as above are, respectively, employed on the multisource data to extract distinct and complementary spatial features. After that, according to the consistency of the same-attribute spatial information extracted from the HSI-LiDAR data in characteristic representation, different attributes of spatial information are combined with spectral–elevational information to depict the land covers from different views. Thus, multiple views can present significantly different information from each other and, meanwhile, reflect distinct and complementary properties of the same-class land covers. Suppose that $\boldsymbol{X}^{Hm}\in\Re^{d_m^{Hspa}\times N}$ and $\boldsymbol{X}^{Lm}\in\Re^{d_m^{Lspa}\times N}$ denote the spatial features extracted from the multisource data, respectively, where d_m^{Hspa} and d_m^{Lspa} are the dimensionality of spatial features for HSI and LiDAR data, respectively, in the mth($m=1,2,$ and 3) view, as indicated in Fig. 2. Given that the physical meaning correlation represented by different attributes of spatial information, the obtained multi-attribute spatial features are fused with the multisource data to form multiview re-expressed HSI-LiDAR data (MVR-HL) with diversity and complementarity. For each view, the re-expressed multisource data could be depicted as

$$\begin{aligned}\boldsymbol{R}_m^H&=[\boldsymbol{H};\boldsymbol{X}^{Hm}]\in\Re^{d_m^H\times N}\\\boldsymbol{R}_m^L&=[\boldsymbol{L};\boldsymbol{X}^{Lm}]\in\Re^{d_m^L\times N}\end{aligned}\tag{5}$$

where $d_m^H=d_{\mathrm{Spe}}+d_m^{Hspa}$ and $d_m^L=d_{\mathrm{Ele}}+d_m^{Lspa}$ represent the dimension of HSI-derived and LiDAR-derived data among the MVR-HL data, respectively. Thus, we explicitly construct the re-expressed spectral–spatial–elevation-fused information form of each view. Each view has its specific and partial information to describe the land covers and different views provide abundant and complementary information representation for all the samples. Finally, the MVR-HL data would be delivered to the MVFL model in the next stage for learning more robust feature representations from multiple views.

B. MB-DCGCNs Model for MVFL

In order to explore view-specific characteristics of each view and capture the intrinsic correlation of the same-class

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land covers, a DCGCNs model that composed of two GCN channels is introduced as the backbone and applied on the re-expressed multisource data of each view to mine informative land cover features. Then, the MB-DCGCN model based on one-view-one-DCGCNs mechanism is designed for learning multiview and collaborative feature representations. In this way, each DCGCN module in the MB-DCGCNs model could extract discriminative spatial–spectral–elevation features and produce independent classification map for each view [44].

For the MVR-HSI data in each view, we construct the graph $\boldsymbol{G}_m^H = [\boldsymbol{V}_m^H, \boldsymbol{A}_m^H]$ to utilize the spectral–spatial–elevation information more comprehensively, where $\boldsymbol{V}_m^H$ is comprised of a series of nodes (vertexes) and $\boldsymbol{A}_m^H$ is an adjacency matrix, which indicates the existence of an edge between each pair of nodes. Since the ijth element in $\boldsymbol{A}_m^H$ denotes the edge weight between nodes $\boldsymbol{V}_{i,m}^H$ and $\boldsymbol{V}_{j,m}^H$, then $\boldsymbol{A}_{ij,m}^H$ can be calculated by the radial basis function (RBF)

$$\boldsymbol{A}_{ij,m}^H = \exp\left(-\left\|\boldsymbol{x}_{i,m}^H - \boldsymbol{x}_{j.m}^H\right\|^2/\sigma^2\right) \tag{6}$$

where (i, j) represents the position in the matrix, m represents the view number, and σ is a parameter to control the width of RBF [35]. The vectors $\boldsymbol{x}_{i,m}^H$ and $\boldsymbol{x}_{j,m}^H$ denote the re-expressed HSI data associated with the vertexes $\boldsymbol{v}_{i,m}^H$ and $\boldsymbol{v}_{j,m}^H$, respectively. Then, the feature vector $\hat{h}_{i,m}^{H\ (l)}$ of each node $\boldsymbol{v}_{i,m}^H$ at the lth graph convolutional layer can be computed as the weighted average of its own information and that of its nearest neighbors

$$\hat{h}_{i,m}^{H\ (l)} = \sum_{j,j\neq i} \hat{\boldsymbol{A}}_{ij,m}^H \hat{h}_{j,m}^{H\ (l-1)} + \hat{h}_{i,m}^{H\ (l-1)} \tag{7}$$

where $\hat{A}_m^H = (\boldsymbol{D}_m^H)^{-1/2}\boldsymbol{A}_m^H(\boldsymbol{D}_m^H)^{1/2}$ is the normalization of the adjacency matrix $\boldsymbol{A}_m^H$ and $\boldsymbol{D}_m^H = \text{diag}(d_{1,m}^H, \ldots, d_{N,m}^H)$ is the degree matrix with its ith element, which denotes the row sum of the adjacency matrix $\boldsymbol{A}_m^H$. To address the numerical instabilities and vanishing gradients in the graph convolution operation, the renormalization terms of $\boldsymbol{A}_m^H$ and $\boldsymbol{D}_m^H$ are introduced as $\tilde{\boldsymbol{A}}_m^H = \boldsymbol{A}_m^H + \boldsymbol{I}$ and $\tilde{\boldsymbol{D}}_{ii,m}^H = \sum_j \tilde{\boldsymbol{A}}_{ij,m}^H$, respectively, to enhance stability in the process of network training. Thus, the feature aggregation of all nodes over the entire graph can be performed by a simple matrix operation, and this process can be expressed as follows:

$$\hat{H}_m^{H(l)} = \left(\tilde{\boldsymbol{D}}_m^H\right)^{-1/2}\tilde{A}_m^H\left(\tilde{\boldsymbol{D}}_m^H\right)^{1/2}\tilde{\boldsymbol{H}}_m^{H(l-1)} \tag{8}$$

where $\tilde{\boldsymbol{H}}_m^{H(l-1)}$ denotes the learned feature map of the $(l-1)$th graph convolution layer with $\tilde{\boldsymbol{H}}_m^{H(0)} = \boldsymbol{R}_m^H$. Hence, for the MVR-HSI data in the mth view, the propagation rule can be formulated as

$$\begin{aligned}\tilde{\boldsymbol{H}}_m^{H(l)} &= \text{ReLU}\left(\hat{\boldsymbol{H}}_m^{H(l)}\boldsymbol{W}^{H(l)} + \boldsymbol{b}^{H(l)}\right)\\ &= \text{ReLU}\left(\left(\tilde{\boldsymbol{D}}_m^H\right)^{-1/2}\tilde{A}_m^H\left(\tilde{\boldsymbol{D}}_m^H\right)^{1/2}\tilde{\boldsymbol{H}}_m^{H(l-1)}\boldsymbol{W}^{H(l)} + \boldsymbol{b}^{H(l)}\right)\end{aligned} \tag{9}$$

where $\tilde{\boldsymbol{H}}_m^{H(l-1)}$ and $\tilde{\boldsymbol{H}}_m^{H(l)}$ are the input and output feature representation of layer l, respectively, and ReLU is the nonlinear activation function with respect to the trainable weights $\{\boldsymbol{W}^{H(l)}\}_{l=1}^p$ and the biases $\{\boldsymbol{b}^{H(l)}\}_{l=1}^p$. Likewise, for the MVR-LiDAR data, the graph structure can be formally expressed as $\boldsymbol{G}_m^L = [\boldsymbol{V}_m^L, \boldsymbol{A}_m^L]$ with $\boldsymbol{V}_m^L$ and $\boldsymbol{A}_m^L$ denoting the sets of nodes and edges, respectively. Mathematically, the output graph representation of the lth layer can be written as

$$\begin{aligned}\tilde{\boldsymbol{H}}_m^{L(l)} &= \text{ReLU}\left(\hat{\boldsymbol{H}}_m^{L(l)}\boldsymbol{W}^{L(l)} + \boldsymbol{b}^{L(l)}\right)\\ &= \text{ReLU}\left(\left(\tilde{\boldsymbol{D}}_m^L\right)^{-1/2}\tilde{A}_m^L\left(\tilde{\boldsymbol{D}}_m^L\right)^{1/2}\boldsymbol{H}_m^{L(l-1)}\boldsymbol{W}^{L(l)} + \boldsymbol{b}^{L(l)}\right).\end{aligned} \tag{10}$$

To improve the identification capacity of land covers, we enhance the collaboration between different channels in each DCGCNs module to learn discriminative spatial–spectral–elevation features from each branch in the MB-DCGCNs model. For each view, the HSI features and LiDAR features extracted by their own respective GCN channel are first fused together in the cascade processing manner. Then, considering that the overfitting problem caused by information redundancy and feature duplication in some multisource classification methods using multiple feature learning and stacking, the concatenated features of these two channels would be further integrated by two FC layers to suppress unimportant features and yield the discriminative feature representations for specific-view land cover category identification. Formally, the final spatial–spectral–elevation features would be sent to the label indicator (e.g., softmax classifier, in our case) for node class prediction, and the output prediction result of the mth view can be expressed as

$$\hat{\boldsymbol{Y}}_m = \text{softmax}\left\{f_{\text{FC2}}\left[f_{\text{FC1}}\left(\tilde{\boldsymbol{H}}_m^H \middle\| \tilde{\boldsymbol{H}}_m^L\right)\right]\right\} \tag{11}$$

where $f_{\text{FC}}(\cdot)$ and $\|$ represent the FC operation and the concatenation operation, respectively. Each DCGCNs model in the MB-DCGCNs model is independently trained using the Adam optimization algorithm with a specific learning rate to capture the graph features of the MVR-HL data from each specific view. The network parameters of each branch could be optimized using the cross-entropy loss as follows:

$$\mathcal{L}\left(\hat{\boldsymbol{Y}}_m^{\text{tr}}, \boldsymbol{Y}^{\text{tr}}\right) = -\sum_i \boldsymbol{Y}_i^{\text{tr}} \log\left(\hat{\boldsymbol{Y}}_{i,m}^{\text{tr}}\right) \tag{12}$$

where $\hat{\boldsymbol{Y}}_{i,m}^{\text{tr}}$ denotes the corresponding prediction class label of the ith testing sample in the mth view. As elaborated above, each DCGCN module that characterizes the intrinsic information exploration of land covers under the same attribute of spatial information is parallelly deployed on the re-expressed multisource to learn the view-specific information and obtain discriminative feature representations of each individual view. Thus, the MB-DCGCNs model could extract different views of spatial–spectral–elevation features from the MVR-HL data to promote the robustness and integrity of land cover representations. Then, in order to comprehensively exploit intrinsic characteristics of each view and the complementarity and consistency among multiple views, the MB-DCGCNs model yields view-specific classification map and the classification maps of multiple views would be further fed into the follow-up land cover label predictor module to produce the unified classification result for subsequent land cover identification tasks.

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C. PHCLA for Land Cover Classification

In the feature-level information fusion architecture, the MB-DCGCNs model that consists of multiple DCGCNs branches is applied on the MVR-HL data, and each branch that could extract spectral–spatial information from HSI and elevation–spatial information from LiDAR data produces its own classification map from a specific view. Given that the complexity of the imaging background and the difference of the gathering condition for the multisource RS data, it is difficult to determine which view is optimal for land cover classification. Different views' information may contribute unequally to discriminate different categories of landcovers; thus, the classification results of multiple views need to be fused properly. Here, we design a novel PHCLA scheme, which consists of decision-level label prediction and pixel-level label assignment to make full use of both the uncertainties within each independent view and consistency among different views for high-confidence classification results [45], [46].

As aforementioned, we obtain ternary classification maps of the MB-DCGCNs model in the previous process. For simplicity, the classification results of the testing samples from each view can be expressed as $\boldsymbol{Y}_m^{\text{te}} \in \Re^{N_{\text{te}} \times C}$ and the class label $y_{i,m}^{\text{te}} \in \{1, C\}$ corresponds to the input testing sample $\boldsymbol{h}_i^{\text{te}} \in \Re^{d_{\text{Spe}} \times N_{\text{te}}}$ in the raw HSI data from the $m^{th}(m = 1, 2, \text{and}3)$ view. Here, the output of each branch is fused using the hard majority voting rule to achieve the optimal prediction result, with the aim of alleviating the problem of misclassification in the single view. For the multisource dataset with N_{te} testing samples, we integrate multiview classification maps to perform label prediction for each testing sample by the following equation:

$$\boldsymbol{Y}_i^{\text{te}} = \left\{ \begin{matrix} \boldsymbol{Y}_{i,1}^{\text{te}} & \text{if } \boldsymbol{Y}_{i,1}^{\text{te}} = \boldsymbol{Y}_{i,2}^{\text{te}} \text{ and } \boldsymbol{Y}_{i,2}^{\text{te}} = \boldsymbol{Y}_{i,3}^{\text{te}} \\ 0 & \text{else} \end{matrix} \right\}. \tag{13}$$

After decision-level information fusion, the testing samples can be divided in two groups depending on the value difference in the predicted label vector $\boldsymbol{Y}^{\text{te}}$, which named as certain samples and uncertain samples. The former samples have the consensus class label of different views, while the latter sample set refers to the samples, where the output results of the MB-DCGCNs are inconsistent in label indicator and can be expressed by the original HSI data as $\boldsymbol{H}^{\text{un}} = \{\boldsymbol{h}_i^{\text{un}}\}_{i=1}^{N_{\text{un}}} \in \Re^{d_{\text{Spe}} \times N_{\text{un}}} (N_{\text{un}} \ll N_{\text{te}})$, which need to be classified by the use of pixel-level information integration in the next stage.

Considering that the same-category land covers are highly concentrated in the RS scene, the land cover distribution information is closely associated with category labels. Specifically, the samples in the local and homogenous region most likely share the same label. Hence, we utilize pixelwise information learning to dig out the underlying relationship among the same-class land covers in the local and homogenous region and complete the class predictions of the uncertain samples. The joint use of the location information, the contour, and edge information, which reflect the spatial distribution of samples with the same label, are performed to reveal pixelwise affinity and local similarity, thereby determining the final class label of each testing sample in the uncertain data pool [43]. First, the gradient information of HSI and LiDAR data, which generated by applying the Sobel operator on the first two PCs of the original HSI and the raw LiDAR, can be expressed as $\boldsymbol{P}$ and $\boldsymbol{Q}$, respectively. Second, we place a $w \times w$ pixel-sized spatial window on each uncertain testing sample $\boldsymbol{h}_i^{\text{un}}$ to identify the local square region. Accordingly, the gradient information of the uncertain sample $\boldsymbol{h}_i^{\text{un}}$ and its spatially adjacent sample set in the local area can be depicted as $[\boldsymbol{P}_i; \boldsymbol{Q}_i]$ and $\boldsymbol{H}^A$, respectively, which reflects the contour and edge characteristics and the local information of the target sample. Then, in order to screen out the samples belong to the same semantic class with the center sample within the same local region, the locally homogeneous sample set $\boldsymbol{H}^H$ of $\boldsymbol{h}_i^{\text{un}}$ can be computed based on the following equation:

$$\boldsymbol{H}^H = \bigcup_j \boldsymbol{h}_j \quad \text{s.t.} \quad \left(\boldsymbol{P}_j = \boldsymbol{P}_i || \boldsymbol{Q}_j = \boldsymbol{Q}_i\right), \quad \boldsymbol{h}_j \in \boldsymbol{H}^A \text{ and } \boldsymbol{h}_j \notin \boldsymbol{H}^{\text{un}}. \tag{14}$$

Suppose there are v samples in $\boldsymbol{H}^H$ with label $\boldsymbol{Y}^H$ and the statistical number of $\boldsymbol{H}^H$ belong to different categories of land covers in the class, which indicates the semantic information distribution of land covers in a local area, can be expressed as

$$\left\{ N^H = \left(n_1^H, n_2^H, \ldots, n_C^H\right) | n_c^H = \sum_{i=1}^{v} \left(\boldsymbol{Y}_i^H == c\right), \quad c = 1, 2, \ldots, C \right\} \tag{15}$$

where n_c^H is the number of samples belong to the class c. Finally, the class label of the center sample $\boldsymbol{h}_i^{\text{un}}$ in the spatial window can be determined by the dominant label information in the locally homogeneous area. Mathematically, the final label of $\boldsymbol{h}_i^{\text{un}}$ can be formulated as

$$\boldsymbol{Y}_i^{\text{un}} = c \quad \text{s.t. } n_c^H = \max(N^H) \text{ and } n_c^H > (v/2). \tag{16}$$

In other words, each uncertain sample would be labeled based on the semantic alignment in the locally homogeneous area. Next, the certain sample pool with the class labels and the uncertain sample set are updated for the next iteration. Then, the above process would be operated on the reduced uncertain sample set under the increasing size of spatial window until there are no available samples for the pixel-level label assignment and the integrated classification map from different views is further processed by the guided filter to preserve the edge and boundary. The underlying assumption is that the samples may keep highly label consistency if they are densely distributed in the locally homogenous region; therefore, we integrate multiple spatial information within the local windowed area of the sample of interest to capture spatial dependencies for final class label identification. Ultimately, the uncertain sample set has been labeled by the use of spatial-context information, and the semantic relatedness of land covers from the same category has been further strengthened to reduce misclassifications in a certain degree.

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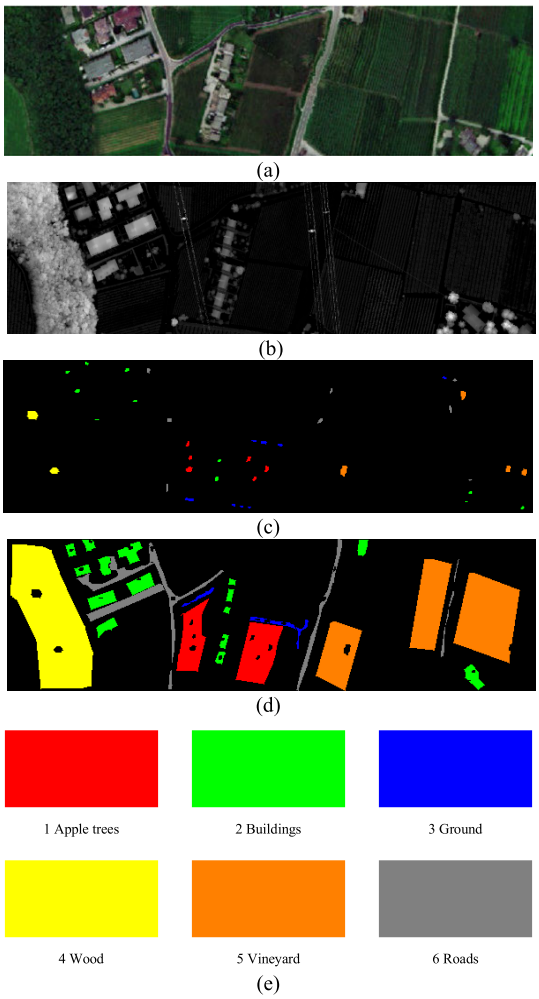


Fig. 3. Visualization of Trento. (a) Pseudo-color image for HSI. (b) LiDAR image. (c) Standard training set in the ground truth map. (d) Standard test set in the ground truth map. (e) Color coding for each class.

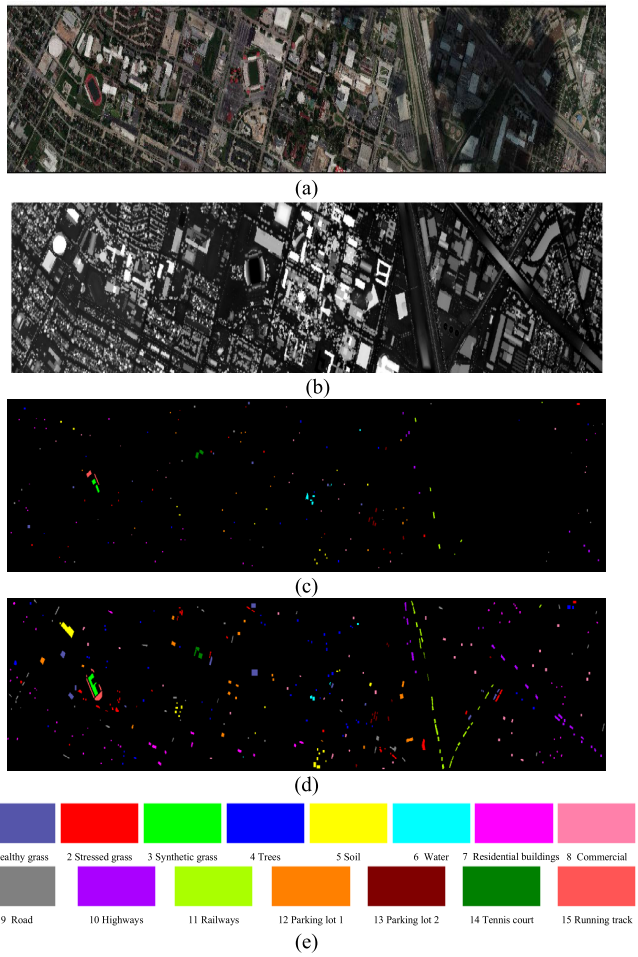


Fig. 4. Visualization of HU. (a) Pseudo-color image for HSI. (b) LiDAR image. (c) Standard training set in the ground truth map. (d) Standard test set in the ground truth map. (e) Color coding for each class.

IV. Experiments and Analysis

A. Experimental Data

In the experiments, to eliminate the influence of randomness by the training set, we validate the performance of the proposed method on two representative datasets, which provide a specific number of training samples in each benchmark dataset. The two representative multisource datasets that collected from different RS sensors are briefly described as follows. The false-color image and the ground truth map of each dataset are illustrated in Figs. 3 and 4, respectively.

1) Trento: The multisource dataset was acquired by the AISA Eagle sensor over Trento, Italy, and covers the spectral range of 420–990 nm with 63 spectral channels at a spatial resolution of 1 m. The scene has 600 × 160 image size with six distinguishable classes.

2) University of Houston 2013 (HU): This image was captured over the University of Houston and its neighboring urban area by the compact airborne spectrographic imager (CASI) sensor. It is comprised of 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral bands ranging from 380 to 1050 nm. The corresponding ground truth image contains 15 classes inside.

The overall experiments are conducted on the standard training set to ensure the fairness, and the corresponding results of each method are averages of five repeated experiments and evaluated by visual comparison and quantitative analysis via the use of four classification metrics, including the class-specific accuracy (CA), overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) quantitatively. Among these metrics, OA records the percentage of correct predictions over the total number of test samples, AA indicates the mean of the percentage of the correct predictions for each class, and kappa provides the overall classwise agreement between the actual class labels (ground truth) and estimated class labels for the testing samples.

B. Comparative Experiments and Discussion

In this section, we investigate the advantages of the proposed method by comparing it with several classical and state-of-the-art classification methods, which are related to MVFL, GCN-based model, multiple-branch CNN structure, and recently developed DL models. These competitive methods can be divided into two categories, i.e., single-source classification algorithms using hyperspectral data (e.g., CNN-MFL [31], DMVL [32], EMCNN [33], and miniGCNs [35])

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TABLE I
CA, OA, AA (%), AND k ($\times 100$) OBTAINED BY DIFFERENT METHODS ON TRENTO

Class code	1	2	3	4	5	6	OA	AA	k
No. Train	129	125	105	154	184	122	--	--	--
No. Test	3905	2778	374	8969	10317	3525	--	--	--
CNN-MFL	80.13	87.53	42.91	99.58	99.33	93.28	93.03	83.79	90.73
DMVL	32.42	44.38	7.98	100	88.04	70.20	60.43	57.17	51.16
EMCNN	100	73.30	73.62	99.99	98.74	98.91	95.76	90.76	94.32
miniGCNs	96.35	77.92	77.60	98.48	99.69	87.52	95.18	89.59	93.54
MFSuDF	89.28	95.16	85.20	99.98	98.67	96.81	97.04	94.18	96.05
MSE	99.49	96.90	100	100	99.49	98.00	99.25	98.98	99.00
FC-Nets	86.05	95.82	65.13	99.91	97.54	97.93	95.85	90.40	94.46
MDL-CNNs	91.30	90.20	60.83	100	100	99.68	97.04	90.34	96.05
CALC	99.74	91.15	87.41	100	99.69	95.28	98.35	95.54	97.79
CCR-Net	88.51	91.93	70.49	100	99.92	98.41	96.81	91.54	95.75
EndNet	85.01	95.62	67.65	99.92	97.44	97.93	95.68	90.60	94.23
GLT-Net	99.69	95.73	95.24	99.98	100	98.49	99.32	98.19	99.09
Proposed	100	100	87.43	100	100	96.43	**99.47**	**97.31**	**99.29**

and multisource data classification methods, including MFSuDF [11], MSE [14], FC-Nets [24], MDL-CNNs [24], CALC [25], CCR-Net [26], EndNet [28], and GLT-Net [29]. For the AP-based methods, such as the CNN-MFL, FC-Nets, MDL-CNNs, and EndNet, the texture information is calculated by the parameter setting exactly as those in the EMAP feature extraction process of our method for a fair performance evaluation. The other comparison methods are implemented with the optimal parameters provided by the open-source library and described in the corresponding literatures.

In the experiments, the details of the proposed method are as follows. For the MVDR process, the parameter setting for extracting multiple spatial features is set the same as those in our previous work [42] based on the theoretical analysis and literature reference. Moreover, the GLCM textures are computed using a series of analysis windows {5, 7, 9, 11, 13} to capture spatial information at different scales. Due to the information dimension difference of the original hyperspectral data, which collected from different types of sensors, both the output dimensions of the graph convolutional layers for the HSI-channel and the FC fusion layer in each branch of the MB-DCGCNs model are set to 128 and 96 for Trento and HU, respectively. Meanwhile, considering that the dimension difference of the multiattribute spatial information, the output dimensions of the graph convolutional layers for the LiDAR-channel is fixed as {16, 32, 32} in order of viewpoint. The initial size of spatial window in the PHCLA scheme is set to 9. For the GCN-related models, the number of nearest neighbors is set to 5. Each model is trained for 300 epochs with the learning rate at 0.001.

Table I lists the details of quantitative results obtained by each classification method on the Trento dataset. The classification maps of all the investigated methods have also been shown in Fig. 5 for intuitive comparison. Above all, the proposed method shows competitive classification performance when compared with the other DL-based classification methods. Concretely, the value of the OA gained by the proposed method is 99.47%, which yields 1.1% and 3.8% improvements with regard to the CALC and Endnet, respectively. It can also be observed that, for the MVFL methods, such as CNN-MFL and EMCNN, our method also gains the improvements with 6.4% and 3.7% in OA, respectively. Furthermore, the classification maps on subregions of the second and sixth categories in the Trento (Roads and Buildings) of each method, which have similar spectrum and are difficult to distinguish, are also given to further evaluate the classification performance. The proposed method achieves an OA of 100% and 96.43% on these two categories, while the mean OAs of the competitive methods are close to 86.30% and 94.37%, respectively. The classification maps, which are shown in Fig. 5, also demonstrate that the proposed method can produce a more accurate classification result map by the enhanced consistency of the same class.

Table II and Fig. 6 report the classification accuracy and classification map of each method on the HU dataset, respectively. Compared with the Trento scene where the landcovers of the same category in the scene are mostly distributed in blocks, the distribution situation of land covers in the HU scene is more complex. From the results, the proposed method is better than the others on three common performance metrics and shows balanced land cover identification for each category on the whole. Fig. 6 also shows the subregion classification results of the seventh and ninth categories in the HU, namely, residential building and road, which are not easily recognized. The per-class accuracy of the other methods on these two categories is basically around 77.35% and 84.48%, and the proposed method can improve the accuracy by nearly 14% points on average. It is worth noting that the proposed method utilizes both the image contour of HSI and that of LiDAR as the base condition for screening the locally homogeneous sample set, which contributes to demarcating the samples of the same category with different spectral reflectances and, hence, yields considerable CAs on the first and second categories, namely, healthy grass and stressed grass, whose spectral information are seriously affected by the cloud cover.

Overall, it is visible that the proposed method outperforms the single-source classification methods (e.g., CNN-MFL and EMCNN) with a significant margin for each dataset, which can be attributed to the fact that the LiDAR data can provide auxiliary elevation information for the HSI data, and the joint use of these two RS data in an effective way could help distinguish different categories of land covers. In this work, all the experiments have been carried out on the Windows 10 desktop computer with 2.9-GHz Intel Core i7-10700 processor, GTX 1080 Ti, and 32-GB memory. Compared to the second-best and recently proposed method GLT, the mean running time increased around 1380 s, while the OA index improved by 1.1% on average for all the datasets. It is also worth noting that the proposed method presents a relatively low requirement for computing device and could be executed on the hardware platforms without GPU support. Among all the three parts in the proposed method, both the MVDR architecture and the MB-DCGCNs model have parallel structure and perform synchronous operation. Moreover, the PHCLA scheme merely deals with the label assignment of only a small number of uncertain samples and, hence, demands not too much computation time. The MVFL process, which is based on the MB-DCGCNs model, plays a leading role in overall operating efficiency and the average running time of this part accounts for over 95% of the total running time.

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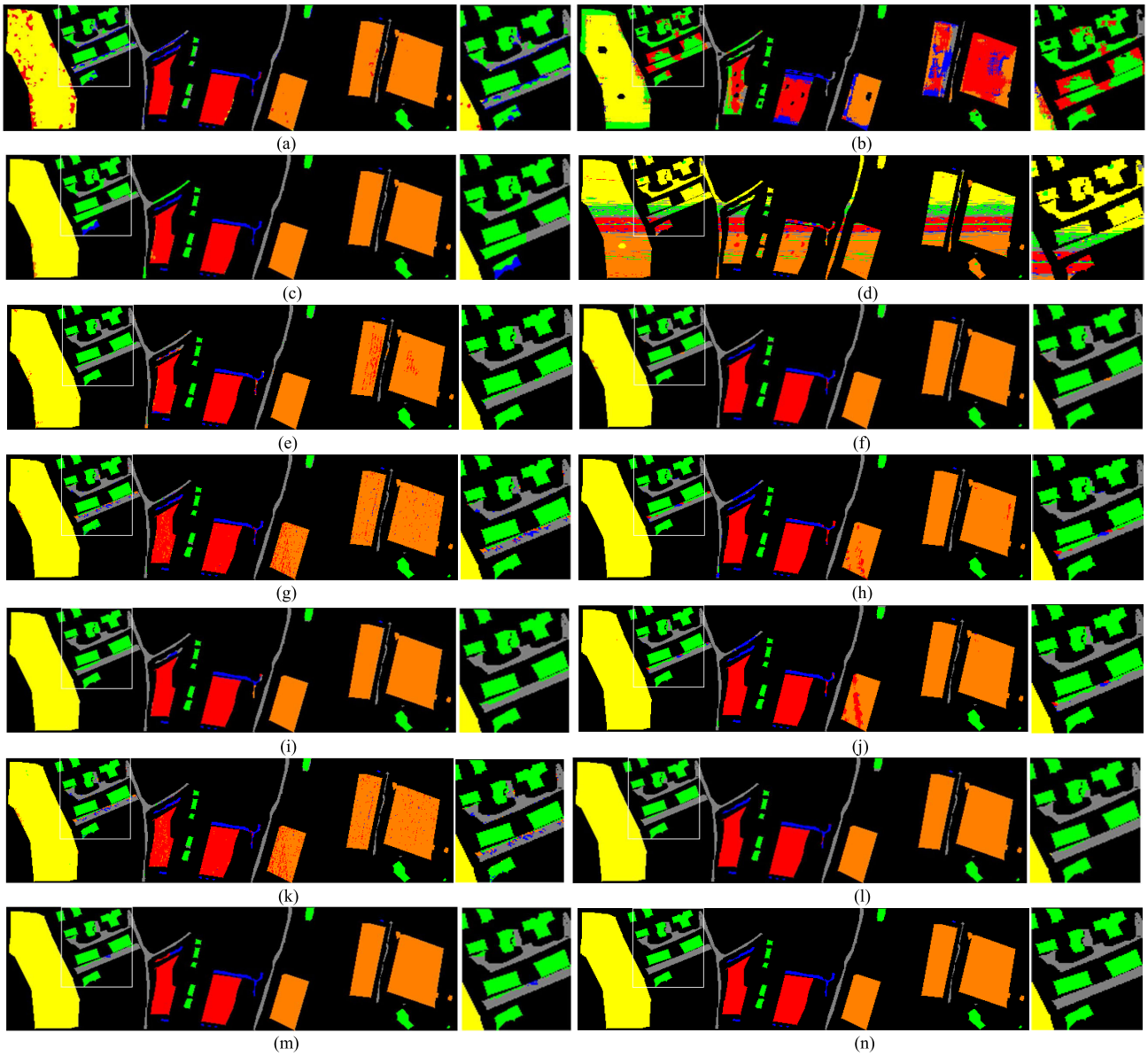


Fig. 5. Classification maps of each method on Trento. (a) CNN-MFL. (b) DMVL. (c) EMCNN. (d) miniGCNs. (e) MFSuDF. (f) MSE. (g) FC-Nets. (h) MDL-CNNs. (i) CALC. (j) CCR-Net. (k) EndNet. (l) GLT-Net. (m) Proposed method. (n) Ground truth.

TABLE II

CA, OA, AA (%), AND k ($\times 100$) OBTAINED BY DIFFERENT METHODS ON HU

Class code	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	OA	AA	K
No. Train	198	190	192	188	186	182	196	191	193	191	181	192	184	181	187	--	--	--
No. Test	1053	1064	505	1056	1056	143	1072	1053	1059	1036	1054	1041	285	247	473	--	--	--
CNN-MFL	98.42	96.96	100	95.05	96.31	18.29	64.77	71.55	80.76	85.36	80.12	75.07	94.18	86.36	99.78	81.22	82.87	79.70
DMVL	66.23	53.52	88.17	62.16	87.00	54.33	69.55	45.50	74.62	65.33	96.36	61.36	87.12	98.39	60.76	68.84	71.36	66.19
EMCNN	83.40	90.74	88.58	90.98	94.65	100	95.34	97.72	79.47	91.60	97.57	90.01	94.98	100	83.37	90.89	91.89	90.11
miniGCNs	99.88	99.23	99.23	99.09	99.15	12.74	64.67	80.20	79.23	74.97	85.75	86.33	71.23	99.09	97.28	80.07	83.20	78.47
MFSuDF	85.24	83.96	41.58	85.29	69.04	30.68	86.02	76.26	83.01	97.39	86.10	81.48	87.84	99.20	97.21	77.52	79.35	79.43
MSE	80.65	96.14	68.61	99.30	96.61	17.28	70.31	88.06	68.44	75.75	87.60	68.25	96.32	80.72	99.37	78.84	79.56	77.11
FC-Nets	98.87	97.91	84.73	98.87	97.21	22.61	72.6	89.55	95.08	98.71	88.62	75.92	84.94	95.37	99.57	86.72	86.7	85.62
MDL-CNNs	99.77	99.01	98.63	100	99.9	17.23	67.68	93.39	90.42	91.98	94.64	93.36	88.61	95.63	100	87.59	88.69	86.58
CALC	97.75	99.67	31.26	98.60	100	90.79	90.65	94.30	98.08	97.96	93.72	88.87	95.17	100	93.48	87.69	91.35	86.68
CCR-Net	99.66	100	65.10	99.90	99.90	19.06	77.19	93.46	89.53	92.63	97.46	91.37	93.02	98.31	99.15	86.95	87.00	85.87
EndNet	81.58	83.65	100	93.09	99.91	95.10	82.65	81.29	88.29	89.00	83.78	90.39	82.46	100	98.10	88.52	89.95	87.59
GLT-Net	100	100	48.35	97.70	100	40.06	86.79	88.52	92.96	96.47	99.88	85.08	98.48	100	94.04	89.03	88.55	88.13
Proposed	83.10	85.15	100	100	100	100	96.64	80.25	92.73	68.53	91.37	100	92.98	100	100	**91.04**	**92.72**	**90.29**

In brief, the comprehensive experiments indicate that the proposed method could boost the classification accuracy with acceptable computational complexity.

C. Ablation Study and Parameter Analysis

1) Module Ablation: As introduced in this article, the proposed method which named after "MVFL-MLIF" for short

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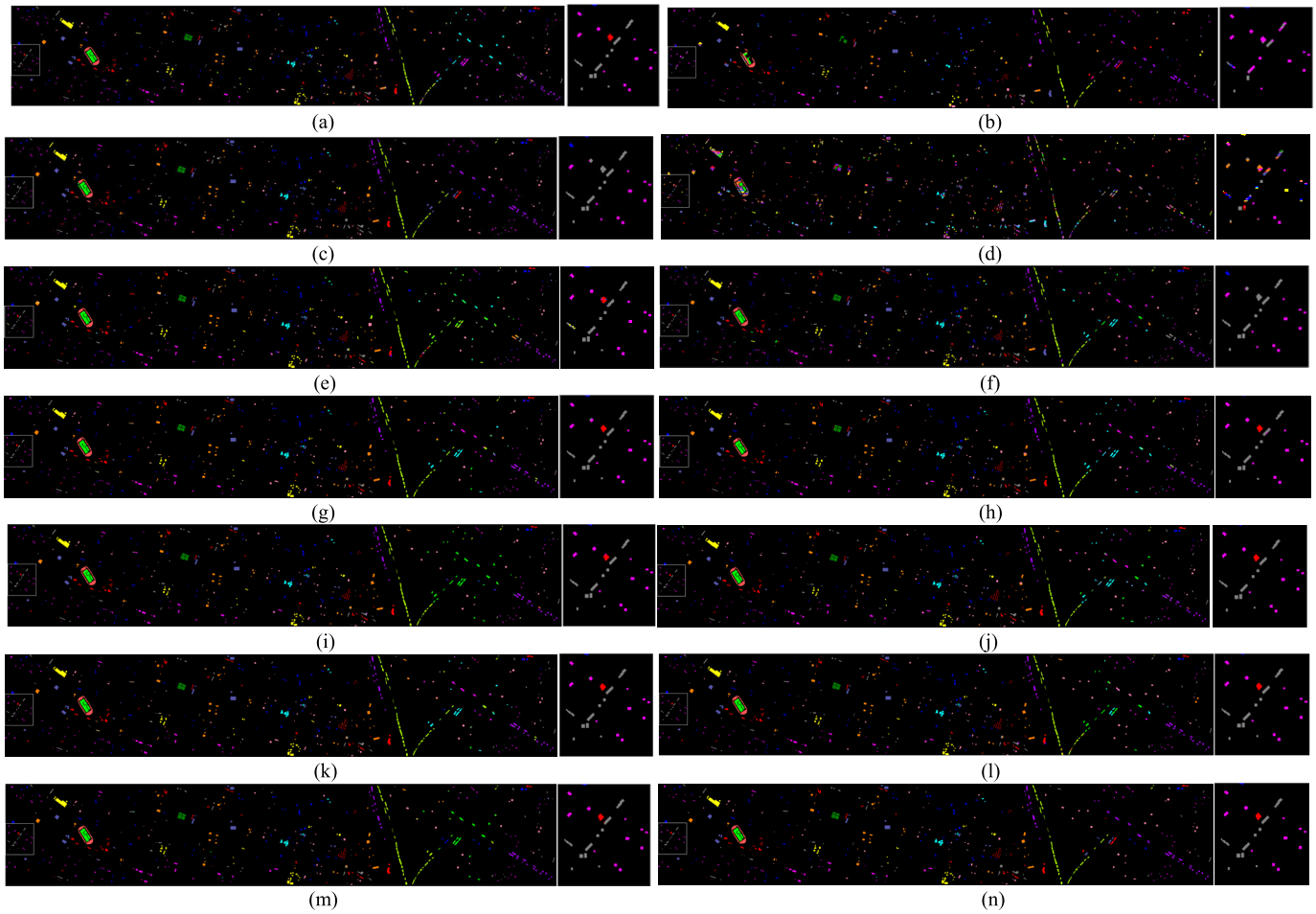


Fig. 6. Classification maps of each method on HU. (a) CNN-MFL. (b) DMVL. (c) EMCNN. (d) miniGCNs. (e) MFSuDF. (f) MSE. (g) FC-Nets. (h) MDL-CNNs. (i) CALC. (j) CCR-Net. (k) EndNet. (l) GLT-Net. (m) Proposed method. (n) Ground truth.

TABLE III
CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD AND ITS VARIANTS

Data set \ Method		View 1	View 2	View 3	View 1,2	View 1,3	View 2,3	MVFL	MVFL-MLIF(HSI)	MVFL-MLIF
Trento	OA	98.03	90.22	95.50	98.20	98.57	97.19	98.72	97.53	99.47
	AA	91.54	79.50	86.04	92.77	92.72	90.01	93.95	95.65	97.31
	k	97.37	87.00	94.02	97.59	98.08	96.23	98.29	96.70	99.29
HU	OA	87.22	79.77	73.84	89.35	86.96	82.98	88.60	86.19	91.03
	AA	88.70	82.57	77.45	90.76	89.46	86.69	91.35	88.75	92.72
	k	86.13	78.13	71.73	88.46	85.86	81.60	87.66	85.07	90.29

mainly consists of three parts, i.e., the MVDR for multisource data, the MB-DCGCNs model, and the PHCLA scheme. In this part, we aim to validate the contribution of each part by a series of ablation experiments. For simplicity, we utilize some abbreviations to represent each model before providing experimental results: "View i" represents the reduced methods that merely reserve single-view feature learning based on the DCGCNs model, "View i, j" represents the reduced methods that reserve double-view feature learning based on two-branch DCGCNs model, and "MVFL" contains the MB-DCGCNs model without the subsequent PHCLA scheme. Meanwhile, the last two models adopt the largest probability average of the possibility output produced by each branch to determine the class label of each unlabeled sample. Moreover, the proposed method has also been modified and applied on the HSI data via the use of multibranch GCNs and the adjusted PHCLA scheme which named after MVFL-MLIF (HSI) for comparison. Table III presents the results of our ablation study on the proposed method for multimodal data fusion and classification with the same parameter configuration as mentioned above. The comparative results including OA, AA, and k are exhibited in Table III.

Above all, it is visible that the MVFL-MLIF scheme outperforms the other comparison models with a significant margin, which can be attributed to the reason that the proposed method utilizes the spatial–spectral–elevation information in the multisource data to capture informative features from different views and enhance the discriminative capability of

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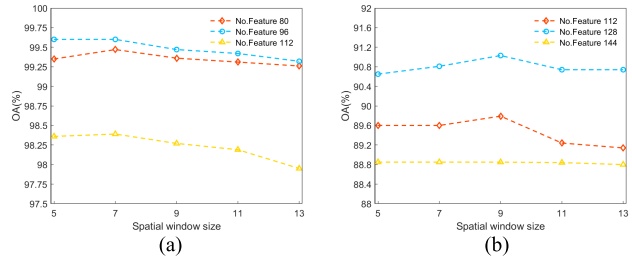


Fig. 7. OA results of the proposed method with varied number of feature and size of initial spatial window on all datasets. (a) Trento. (b) HU.

land covers by MLIF. Compared to the single-view models, both the double-view models and the MVFL model aim to extract features from different views for addressing insufficiency in each individual view. However, it can be observed that the OAs do not always increase with the augmentation of view-specific information. This is because the fusion scheme for different views in these two models has not taken the view-specific intrinsic information and the relationships among different view into consideration, hence, could not guarantee the outperformance of land cover classification accuracy over their corresponding single-view models. Aiming to jointly use the advantages of view-specific peculiar representation and the complementarity and consistency among different views, the proposed method first exploits the latent consistency of classification results from multiple views and integrates the classification results from multiple views into a unified classification map by the decision-level information fusion. Then, the pixel-level label assignment scheme would be further performed to capture the pixelwise correlations and yield the final classification map by the use of semantic information alignment in the local region with homogeneity. Hence, the proposed method could achieve better performance than the best of single-view and dual-view comparison methods. Meanwhile, the inferior performance of the MVFL model in contrast with the MVFL-MLIF model has also validated the effectiveness of the PHCLA scheme in the various and heterogenous information fusion for multisource data classification from the side. Furthermore, the proposed method improves the OA over 1.9% and 4.8% for Trento and HU, respectively, in contrast with the MVFL-MLIF scheme using hyperspectral data alone owing to the introduction of LiDAR data, which also proves that the complementary elevation information plays an important role in the improvement of classification accuracy. All in all, the classification accuracy of the proposed method continues to rise with the introduction of each part, which directly proves the rationality and effectiveness of each part in terms of multisource data classification performance improvement.

2) Parameter Analysis: In the proposed method, there are several crucial parameters that impact the classification performance, mainly including the feature dimension in the MVFL procedure and the window size in the PHCLA scheme. Thus, the impact of the feature dimension and window size on the proposed method has been studied, and the corresponding results are plotted in Fig. 7. It can be observed that proper feature dimension of each dataset corresponds to the best classification accuracy, which accords with the fact that the large-dimensional feature easily leads to outfitting and the small-dimensional feature could not effectively distinguish different categories of land covers under the fixed learning rate and iteration number. Hence, the appropriate configurations would be beneficial to the MB-DCGCNs model in learning robust and discriminative feature representations and further promote the classification accuracy. Moreover, the OA result at first rises as the initial window size increases and then decreases. The reason is that when the window size is small, the contextual information around the uncertain sample of interest would be not explored fully to predict the class label of the local homogenous area. As the window size increases succeedingly, the multisource information inherent in the local area becomes richer, which leads to the reliability improvement of the predicted class label. However, once a certain size is reached, further expansion of the window size will introduce additional interference samples, which would deteriorate the classification performance. Moreover, compared to the HU data, the Trento data have larger and more densely distributed land covers to discriminate, hence, tend to achieve high classification accuracy with relatively small-size spatial window.

V. Conclusion

In this article, we proposed a novel multisource classification method, which mainly consists of three parts. In the process of multisource data re-expression, multiattribute spatial information with otherness and complementarity, which implicitly embedded in the multisource data, are deployed to reconstruct the data presentation form for avoiding information loss in a certain degree and facilitating the follow-up feature extraction procedure. In the feature-level information fusion process for HSI-LiDAR data, an MB-DCGCNs model is developed and employed on the re-expressed multisource RS data for extracting feature representations from different views of point; this enables better spectral–elevation–spatial information integration of each specific view independently and individually. In the end, a progressively high-reliable label assignment scheme combined by decision-level label prediction and pixel-level label assignment is designed to comprehensively exploit the complementarity and consistency among multiple views and the pixelwise dependency relationship of the locally homogeneous samples for yielding the final classification result. Experimental results on real-world datasets demonstrate that the proposed method exhibits competitive performance with respect to some state-of-the-art methods on the joint classification of HSI and LiDAR data. In addition, the ablation experiments have indicated the effectiveness of each part in the proposed method.

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